

Overview of Statistical Learning and Modeling – 36-600

Lab 7R: Random Effects
Due Friday, October 10, 11:59pm

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Today's lab will be relatively short, so that you can turn it in and head off to enjoy fall break.

Data

Here we read in the sleep study data you used in Project 1 and we retain three variables: `cohort` and `term_gpa` as predictors, and `TotalSleepTime` (renamed `sleep`) as the response.

```
suppressMessages(library(tidyverse))
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
df      <- read.csv(url("https://raw.githubusercontent.com/FSKoerner/F25-36600-data/refs/heads/main/c"))
df$cohort <- factor(df$cohort)
df %>% select(., cohort, TotalSleepTime, term_gpa) -> df
names(df)[2] <- "sleep"
summary(df)
```

##	cohort	sleep	term_gpa
##	lac1:131	Min. :194.8	Min. :0.350
##	lac2: 77	1st Qu.:366.9	1st Qu.:3.233
##	nh :147	Median :400.4	Median :3.556
##	uw1 :140	Mean :397.3	Mean :3.450
##	uw2 :139	3rd Qu.:430.1	3rd Qu.:3.810
##		Max. :587.7	Max. :4.000

Note that for this lab, we are not going to split the data into training and test sets.

Question 1

Let's first assume that we regress `sleep` upon `cohort`. Since the cohorts that appear in the study are a randomly selected subset of all possible cohorts, learning a random effects model is appropriate (as opposed to a one-way ANOVA model). Fit an appropriate model using `lmer()` and pass the output to `summary()`. Display the five intercepts for the five cohorts. (*Hint*: use one of the functions shown in the notes, specifically one used with the Palmer Penguin data.) Last, what is the within-cohort correlation? (This last part involves extracting numbers from the `summary()` output and “doing math” as opposed to making further function calls.) Is this a weak, medium, or strong correlation? (If you don't think you have a good sense for that strength, consider using the internet to help you with that judgement - find some examples of the strength of correlation in real data settings that you have a good handle on - and bear in mind that rarely will two tables mapping correlation coefficients to adjectives match.)

```
suppressMessages(library(lme4))
```

```
## Warning: package 'lme4' was built under R version 4.4.3
```

```
model <- lmer(sleep ~ 1 + (1 | cohort), data = df)
summary(model)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: sleep ~ 1 + (1 | cohort)
## Data: df
##
## REML criterion at convergence: 6747.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.3441 -0.6160  0.0175  0.6454  3.7001
##
## Random effects:
## Groups Name Variance Std.Dev.
## cohort (Intercept) 205.7 14.34
## Residual 2429.7 49.29
## Number of obs: 634, groups: cohort, 5
##
## Fixed effects:
## Estimate Std. Error t value
## (Intercept) 395.93 6.72 58.91
```

```
coef(model)$cohort
```

```
## (Intercept)
## lac1 391.9559
## lac2 381.1778
## nh 385.9503
## uw1 411.6616
## uw2 408.9144
```

```
vc <- as.data.frame(VarCorr(model))
var_cohort <- vc[vc$grp == "cohort", "vcov"]
var_resid <- vc[vc$grp == "Residual", "vcov"]

rho <- var_cohort / (var_cohort + var_resid)
rho
```

```
## [1] 0.07803688
```

Given rho is approximately 0.078, we can say that there is a weak correlation.

Question 2

Now let's look at `term_gpa`. Learn a simple linear regression model that has `sleep` as the response variable and `term_gpa` as the predictor variable, and overlay the model on a plot of `sleep` versus `term_gpa`. Also, determine the AIC value for the model.

```
lm_gpa <- lm(sleep ~ term_gpa, data = df)
summary(lm_gpa)
```

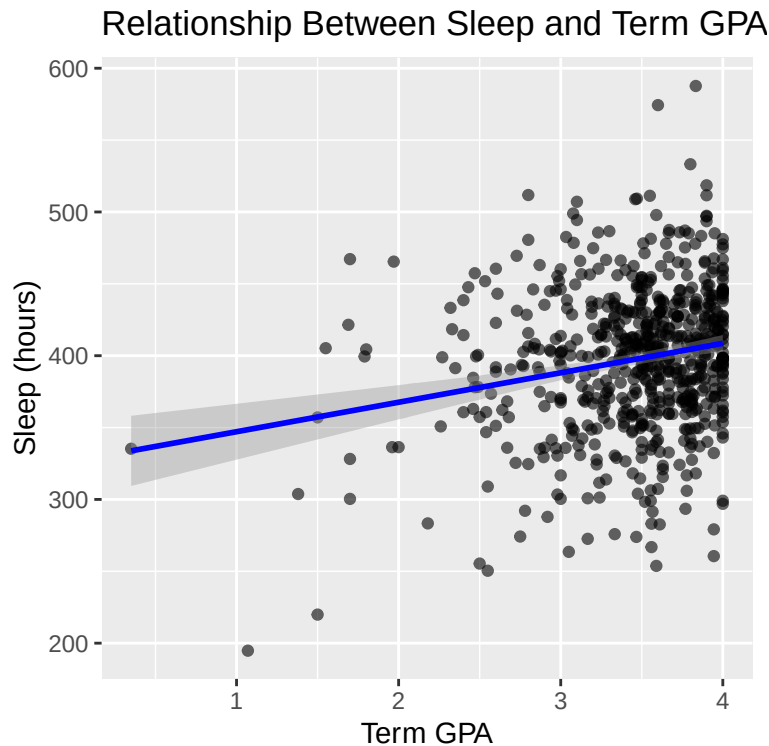
```
##
## Call:
## lm(formula = sleep ~ term_gpa, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -153.773  -31.975    3.135   31.587  182.478
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  326.627     13.800   23.668 < 2e-16 ***
## term_gpa      20.494       3.959    5.176 3.04e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 49.85 on 632 degrees of freedom
## Multiple R-squared:  0.04067,    Adjusted R-squared:  0.03916
## F-statistic: 26.8 on 1 and 632 DF,  p-value: 3.043e-07
```

```
aic_value <- AIC(lm_gpa)
print(aic_value)
```

```
## [1] 6759.875
```

```
ggplot(df, aes(x = term_gpa, y = sleep)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE, color = "blue", linewidth = 1) +
  labs(
    title = "Relationship Between Sleep and Term GPA",
    x = "Term GPA",
    y = "Sleep (hours)"
  )
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Question 3

Now implement a mixed effects model with `term_gpa` as the fixed effect and `cohort` as the random effect, with random intercepts (but sharing the same slope). Output the linear model coefficients for each cohort. (Mentally confirm that indeed, the intercepts are different but the slopes are the same.) And, as you did in Question 2, record the AIC value.

No plot is necessary, for now.

```
mixed1 <- lmer(sleep ~ term_gpa + (1 | cohort), data = df)
summary(mixed1)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: sleep ~ term_gpa + (1 | cohort)
##   Data: df
##
## REML criterion at convergence: 6703.9
##
## Scaled residuals:
##   Min       1Q   Median       3Q      Max
## -3.3027 -0.6417  0.0008  0.6684  3.6546
##
## Random effects:
##   Groups   Name                Variance Std.Dev.
##   cohort   (Intercept)         262.2    16.19
##   Residual                        2285.5    47.81
## Number of obs: 634, groups:  cohort, 5
##
```

```
## Fixed effects:
##           Estimate Std. Error t value
## (Intercept)  309.790     15.557  19.914
## term_gpa      24.971      3.953   6.317
##
## Correlation of Fixed Effects:
##           (Intr)
## term_gpa -0.876
```

```
coef(mixed1)$cohort
```

```
##           (Intercept) term_gpa
## lac1      309.7316 24.97128
## lac2      293.8272 24.97128
## nh        294.5427 24.97128
## uw1       324.8917 24.97128
## uw2       325.9558 24.97128
```

```
aic_mixed1 <-AIC(mixed1)
```

Question 4

Repeat Question 3, now allowing random slopes but the same intercept.

```
mixed2 <- lmer(sleep ~ term_gpa + (0+term_gpa | cohort), data = df)
summary(mixed2)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: sleep ~ term_gpa + (0 + term_gpa | cohort)
## Data: df
##
## REML criterion at convergence: 6705.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.1467 -0.6384  0.0161  0.6801  3.6836
##
## Random effects:
## Groups   Name      Variance Std.Dev.
## cohort  term_gpa    20.05    4.478
## Residual                2293.67  47.892
## Number of obs: 634, groups: cohort, 5
##
## Fixed effects:
##           Estimate Std. Error t value
## (Intercept)  315.424     13.575  23.236
## term_gpa      23.436      4.387   5.342
##
## Correlation of Fixed Effects:
##           (Intr)
## term_gpa -0.881
```

```
coef(mixed2)$cohort
```

```
##      (Intercept) term_gpa
## lac1      315.4243 22.91726
## lac2      315.4243 19.08337
## nh        315.4243 19.45821
## uw1       315.4243 27.62375
## uw2       315.4243 28.09606
```

```
aic_mixed2 <-AIC(mixed2)
```

Question 5

Repeat Question 3, but now with random slopes *and* intercepts.

```
mixed3 <- lmer(sleep ~ term_gpa + (term_gpa | cohort), data = df)
summary(mixed3)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: sleep ~ term_gpa + (term_gpa | cohort)
## Data: df
##
## REML criterion at convergence: 6702
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.3004 -0.6271 -0.0078  0.6573  3.6963
##
## Random effects:
## Groups Name Variance Std.Dev. Corr
## cohort (Intercept) 2668.7  51.66
##        term_gpa    128.4  11.33  -0.97
## Residual      2267.2  47.62
## Number of obs: 634, groups: cohort, 5
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)  296.802    27.484    10.80
## term_gpa      28.456     6.602     4.31
##
## Correlation of Fixed Effects:
##      (Intr)
## term_gpa -0.967
```

```
coef(mixed3)$cohort
```

```
##      (Intercept) term_gpa
## lac1      332.4752 18.29695
## lac2      261.7015 34.15287
## nh        234.3221 41.29623
## uw1       330.5430 23.33440
## uw2       324.9693 25.19796
```

```
aic_mixed3 <-AIC(mixed3)
```

Question 6

Select the best model from among those in Questions 2-5 (i.e., the one with the lowest AIC value). Plot the data again, and overlay the best model; note that this may require you to make five separate calls to `geom_abline()`. *Note:* to access the slopes and intercepts, you could use, e.g.,

```
coefs <- coef(lmer.out)$cohort
coefs[1, 1]  # first row, first column -- intercept of first cohort
coefs[3, 2]  # third row, second column -- slope of third cohort
etc.
```

In the end, your plot may not clearly show five separate lines, which is fine, because the `lac1` and `lac2`, and `uw1` and `uw2`, cohorts had very similar data.

```
#print(aic_value)
#print(aic_mixed1)# <- best
#print(aic_mixed2)
#print(aic_mixed3)

best_model <- mixed1
coefs <- coef(best_model)$cohort

p <- ggplot(df, aes(x = term_gpa, y = sleep, color = cohort)) +
  geom_point()

# Add regression lines for each cohort
p <- p +
  geom_abline(intercept = coefs["lac1", 1], slope = coefs["lac1", 2], color = "red") +
  geom_abline(intercept = coefs["lac2", 1], slope = coefs["lac2", 2], color = "blue") +
  geom_abline(intercept = coefs["uw1", 1], slope = coefs["uw1", 2], color = "green") +
  geom_abline(intercept = coefs["uw2", 1], slope = coefs["uw2", 2], color = "purple") +
  geom_abline(intercept = coefs["other", 1], slope = coefs["other", 2], color = "orange") +
  theme_minimal() +
  labs(title = "Term GPA vs Outcome with Best Model Lines",
       x = "Term GPA",
       y = "sleep")
print(p)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_abline()').
```

